5

Density functional theory

5.1 Introduction

In the previous chapter we saw how the many-electron problem can be treated in the Hartree–Fock formalism in which the solution of the many-body Schrödinger equation is written in the form of a Slater determinant. The resulting HF equations depend on the occupied electron orbitals, which enter these equations in a nonlocal way. The nonlocal potential of Hartree–Fock is difficult to apply in extended systems, and for this reason there have been relatively few applications to solids; see however Ref. [1].

Most electronic structure calculations for solids are based on density functional theory (DFT), which results from the work of Hohenberg, Kohn and Sham [2, 3]. This approach has also become popular for atoms and molecules. In the density functional theory, the electronic orbitals are solutions to a Schrödinger equation which depends on the electron density rather than on the individual electron orbitals. However, the dependence of the one-particle Hamiltonian on this density is in principle nonlocal. Often, this Hamiltonian is taken to depend on the *local* value of the density only – this is the *local density approximation* (LDA). In the vast majority of DFT electronic structure calculations for solids, this approximation is adopted. It is, however, also applied to atomic and molecular systems [4].

In this chapter we describe the density functional method for electronic structure calculations. In the present section, the physical interpretation of the density functional equations is first described and the formal derivations are given. In the next section the local density approximation is considered. An application to the ground state of the helium atom will be described in some detail in Section 5.5. Finally, some results obtained using density functional theory will be discussed in Section 5.5.3. For further reading, there are many reviews and books available; see for example Refs. [4–7].

5.1.1 Density functional theory: physical picture

In density functional theory, an effective independent-particle Hamiltonian is arrived at, leading to the following Schrödinger equation for one-electron spin-orbitals:

$$\left[-\frac{1}{2}\nabla^2 - \sum_n \frac{Z_n}{|\mathbf{r}-\mathbf{R}_n|} + \int \mathrm{d}^3r'\ n(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|} + V_{\mathrm{xc}}[n](\mathbf{r})\right]\psi_k(\mathbf{r}) = \varepsilon_k\psi_k(\mathbf{r}). \tag{5.1}$$

The first three terms in the left hand side of this equation are exactly the same as those of Hartree–Fock, Eq. (4.31), namely the kinetic energy, the electrostatic interaction between the electrons and the nuclei, and the electrostatic energy of the electron in the field generated by the total electron density $n(\mathbf{r})$. The fourth term contains the many-body effects, lumped together in an exchange-correlation potential. The main result of density functional theory is that there exists a form of this potential, depending only on the electron density $n(\mathbf{r})$, that yields the *exact* ground state energy and density. Unfortunately, this exact form is not known, but there exist several approximations to it, as we shall see in Sections 5.2 and 5.3. The dependence of the independent-particle Hamiltonian on the density only is in striking contrast with Hartree–Fock theory, where the Hamiltonian depends on the individual orbitals. The solutions of Eq. (5.1) must be self-consistent in the density, which is given by

$$n(\mathbf{r}) = \sum_{k=1}^{N} |\psi_k(\mathbf{r})|^2, \tag{5.2}$$

where the sum is over the N spin-orbitals ψ_k having the lowest eigenvalues ε_k in (5.1), and N is the number of electrons in the system.

The total energy of the many-electron system is given by

$$E = \sum_{k=1}^{N}\varepsilon_k - \frac{1}{2}\int \mathrm{d}^3r\ \mathrm{d}^3r'\ n(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}n(\mathbf{r}') + E_{\mathrm{xc}}[n] - \int \mathrm{d}^3r\ V_{\mathrm{xc}}[n](\mathbf{r})n(\mathbf{r}) \tag{5.3}$$

where the parameters ε_k are the eigenvalues occurring in Eq. (5.1) and E_{xc} is the exchange correlation energy. The exchange correlation potential $V_{\mathrm{xc}}[n]$ which occurs in (5.1) is the functional derivative of this energy with respect to the density:

$$V_{\mathrm{xc}}[n](\mathbf{r}) = \frac{\delta}{\delta n(\mathbf{r})}E_{\mathrm{xc}}[n]. \tag{5.4}$$

Although the energy parameters ε_k are not, strictly speaking, one-electron energies they are often treated as such for comparison with spectroscopy experiments according to an extended version of Koopman's theorem (see Problem 5.4). The wave functions ψ_k also have no individual meaning but are used to construct the total charge density. This again contrasts with Hartree–Fock where the one-electron spin-orbitals have a definite interpretation: they are the constituents of the many-electron wave function.

Equations (5.1) and (5.2) are solved in an iterative self-consistency loop, which is started by choosing an initial density $n(\mathbf{r})$, constructing the Schrödinger equation (5.1) from it, solving the latter and calculating the resulting density from (5.2). Then a new Schrödinger equation is constructed and so on, until the density does not change appreciably any more.

In both DFT and Hartree–Fock theory, the electrons move in a background composed of the Hartree and external potentials. In addition to this, the exchange term in Hartree–Fock accounts for the fact that electrons with parallel spin avoid each other as a result of the exclusion principle (exchange hole). Opposite spin pairs do not feel this interaction. In DFT, the exchange-correlation potential includes not only exchange effects but also correlation effects due to the Coulomb repulsion between the electrons (*dynamic correlation* effects). In HF, the exchange interaction is treated exactly, but dynamic correlations are neglected. DFT is in principle exact, but we do not know the exact form of the exchange correlation potential – both exchange and dynamic correlation effects are in practice treated approximately.

It is essential that the exchange correlation energy is given in terms of the electron density only and contains no explicit dependence on the external potential (in our case the potential due to the atomic nuclei). As we shall see in Section 5.2, a local approximation for the exchange correlation energy occuring in the DFT equation (5.1) is usually made, thereby simplifying the implementation significantly with respect to Hartree–Fock with its complicated nonlocal exchange term.

*5.1.2 Density functional formalism and derivation of the Kohn–Sham equations

For a many-electron system, the Hamiltonian is given by

$$H = \sum_i \left[-\frac{1}{2}\nabla_i^2 + V_{\text{ext}}(\mathbf{r}_i) \right] + \frac{1}{2} \sum_{\substack{i,j;\\ i \neq j}} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|}. \tag{5.5}$$

V_{ext} is an external potential which, in the systems of interest to us, is the Coulomb attraction by the static nuclei.

In Chapter 3 we have seen how the ground state can be found by varying the energy-functional with respect to the wave function. Now consider carrying out this variational procedure in two stages: first – for a given electron density – minimise with respect to the wave functions consistent with this density, and then minimise with respect to the density. Denoting by $\min_{\Psi|n}$ a minimisation with respect to the wave functions Ψ which are consistent with the density $n(\mathbf{r})$, we can write

$$E[n] = \min_{\Psi|n} \langle \Psi | H | \Psi \rangle \tag{5.6}$$

and it will be clear that the ground state of the many-electron Hamiltonian can be found by minimising the functional $E[n]$ with respect to the density, subject to the constraint

$$\int \mathrm{d}^3 r \; n(\mathbf{r}) = N \tag{5.7}$$

where N is the total number of electrons.

Now consider a separation of the Hamiltonian into the Hamiltonian H_0 of the homogeneous electron gas (with external potential $V_{\text{ext}} \equiv 0$), and the external potential:

$$H = H_0 + V_{\text{ext}}(\mathbf{r}). \tag{5.8}$$

In this case we can write $E[n]$ as

$$E[n] = \min_{\Psi|n} \left[\langle \Psi | H_0 | \Psi \rangle + \int \mathrm{d}^3 r \; V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \right]. \tag{5.9}$$

If we minimise the term in square brackets for a given density $n(\mathbf{r})$, the second term is a constant so that we do not have to include it in the minimisation:

$$E[n] = \min_{\Psi|n} \left[\langle \Psi | H_0 | \Psi \rangle \right] + \int \mathrm{d}^3 r \; V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}). \tag{5.10}$$

Writing

$$F[n] = \min_{\Psi|n} \left[\langle \Psi | H_0 | \Psi \rangle \right] \tag{5.11}$$

we see that $E[n]$ can be written as

$$E[n] = F[n] + \int \mathrm{d}^3 r \; V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \tag{5.12}$$

and $F[n]$ obviously does not depend on the external potential. We shall now use these general statements to treat our problem of interacting electrons in an external potential. Summarising the results obtained so far, we see that:

- The ground state density can be obtained by minimising the energy-functional (5.6).
- If we split the Hamiltonian H into a homogeneous one, H_0, and the external potential, the energy-functional can be split into a part $F[n]$, which is defined in (5.11) and which is independent of the external potential, and the functional $\int \mathrm{d}^3 r \; V_{\text{ext}}(\mathbf{r}) n(\mathbf{r})$.

The problem with treating the many-electron system lies in the electron–electron interaction. In fact, for both interacting and noninteracting electron systems the form of the functional $E[n]$ is unknown, but the ground state energy for noninteracting electrons can be solved for trivially, and we can use this to tackle the problem of interacting electrons. In the noninteracting case, $E[n]$ has a kinetic contribution and a contribution from the external potential V_{ext}:

$$E[n] = T[n] + \int d^3r\, n(\mathbf{r})V_{\text{ext}}(\mathbf{r}). \tag{5.13}$$

Variation of E with respect to the density leads to the following equation:

$$\frac{\delta T[n]}{\delta n(\mathbf{r})} + V_{\text{ext}}(\mathbf{r}) = \lambda n(\mathbf{r}), \tag{5.14}$$

where λ is the Lagrange parameter associated with the restriction of the density to yield the correct total number of electrons, N. The form of $T[n]$ is unknown, but we know that the ground state of the system can be written as a Slater determinant with spin-orbitals satisfying the single-particle Schrödinger equation:

$$\left[-\tfrac{1}{2}\nabla^2 + V_{\text{ext}}(\mathbf{r})\right]\psi_k(\mathbf{r}) = \varepsilon_k\psi_k(\mathbf{r}). \tag{5.15}$$

The ground state density is then given by

$$n(\mathbf{r}) = \sum_{k=1}^{N} |\psi_k(\mathbf{r})|^2 \tag{5.16}$$

where the spin-orbitals ψ_k are supposed to be normalised so that the density satisfies the correct normalisation to the number of particles N. Using the above analysis, and taking $T[n]$ for the functional $F[n]$, we immediately see that the kinetic energy-functional T is independent of the potential V_{ext}. Summarising, we have:

- The energy-functional of a noninteracting electron gas can be split into a kinetic functional $T[n]$, and a functional representing the interaction with the external potential, $\int d^3r\, V_{\text{ext}}(\mathbf{r})n(\mathbf{r})$. The kinetic functional does not depend on the external potential.
- The exact solution of the noninteracting electron gas is given in terms of the eigenfunction solutions of the independent-particle Hamiltonian; see Eq. (5.15).

The energy-functional for a many-electron system with electronic interactions included can be written in the form

$$\begin{aligned} E[n] = T[n] &+ \int d^3r\, n(\mathbf{r})V_{\text{ext}}(\mathbf{r}) \\ &+ \frac{1}{2}\int d^3r \int d^3r'\, n(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|}n(\mathbf{r}) + E_{\text{xc}}[n], \end{aligned} \tag{5.17}$$

where the last term, the exchange correlation energy, contains, by definition, all the contributions not taken into account by the first three terms which represent the kinetic energy-functional of the *noninteracting* electron gas, the external and the

Hartree energy respectively. It is important to note that we have made no approximations so far but moved all the unknown correlations into E_{xc}, which depends on the density n rather than on the explicit form of the wave function because all the other terms in (5.17) depend on the density. For the interacting electron gas it is not clear that the kinetic energy and the electron–electron interaction can be written as a sum of two terms depending on the density only; therefore the kinetic functional for *noninteracting* electrons, which depends only on the density, has been split off and the remaining part of the kinetic energy has been moved into E_{xc}. Varying this equation with respect to the density, we obtain

$$\frac{\delta T[n]}{\delta n(\mathbf{r})} + \frac{\delta E_{xc}[n]}{\delta n(\mathbf{r})} + \int d^3r'\, n(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|} + V_{ext}(\mathbf{r}) = \lambda n(\mathbf{r}). \tag{5.18}$$

This equation has the same form as (5.14), the only difference being the potential replaced by a more complicated one, the 'effective potential':

$$V_{eff}(\mathbf{r}) = V(\mathbf{r}) + \frac{\delta E_{xc}[n]}{\delta n(\mathbf{r})} + \int d^3r'\, n(\mathbf{r}')\frac{1}{|\mathbf{r}-\mathbf{r}'|}. \tag{5.19}$$

The analogue of Eq. (5.15) now becomes

$$\left[-\frac{1}{2}\nabla^2 + V_{eff}(\mathbf{r})\right]\psi_k(\mathbf{r}) = \varepsilon_k\psi_k(\mathbf{r}). \tag{5.20}$$

Comparing Eqs. (5.20) and (5.17), we see that adding the eigenvalues ε_k of the occupied states does not lead to the total energy as the Hartree energy is overestimated by a factor of 2, and there is a further difference in the exchange correlation term, so that we have altogether:

$$E = \sum_{k=1}^{N}\varepsilon_k - \frac{1}{2}\int d^3r\, d^3r'\, n(\mathbf{r})\frac{1}{|\mathbf{r}-\mathbf{r}'|}n(\mathbf{r}') + E_{xc}[n] - \int d^3r\, V_{xc}[n(\mathbf{r})]n(\mathbf{r}). \tag{5.21}$$

where V_{xc} is defined in (5.4). The density functional procedure is now given by Eqs. (5.16), (5.19), (5.20) and (5.21). These equations were first derived by Kohn and Sham [3].

We have already mentioned that the exact form of the exchange correlation potential is not known. This energy is a functional of the density, but there may be an additional explicit dependence on the external potential. Such a dependence would imply that each physical system has its own particular exchange correlation energy-functional. That the exchange correlation potential does not have such a dependence follows immediately from the argument given at the beginning of this section (Eqs. (5.8–5.12)) by separating the external potential off the Hamiltonian and taking the remaining contributions to the energy-functional for $F[n]$. This shows that the exact exchange correlation potential, which should work for *all* materials, is simply a functional of the density. In practice we have to use approximations for E_{xc}, as the exact form of this functional is unknown, and our approximation might be better for some materials than for others. The final conclusion can then be formulated as follows:

- If we split the energy-functional according to (5.17), the term $E_{xc}[n]$ into which we have moved all the terms we do not have under control, is independent of the external potential.
- The minimisation problem of the energy-functional can be carried out using the Kohn-Sham equations (5.20) together with the constraint (5.16).

5.2 The local density approximation

The difference between the Hartree–Fock and density functional approximation is the replacement of the HF exchange term by the exchange correlation energy E_{xc} which is a functional of the density. The exchange correlation potential is a functional derivative of the exchange correlation energy with respect to the local density and for a homogeneous electron gas this will depend on the value of the electron density. For a nonhomogeneous system, the value of the exchange correlation potential at the point $\mathbf{r}$ depends not only on the value of the density at $\mathbf{r}$ but also on its variation close to $\mathbf{r}$, and it can therefore be written as an expansion in the gradients to arbitrary order of the density:

$$V_{xc}[n](\mathbf{r}) = V_{xc}[n(\mathbf{r}), \nabla n(\mathbf{r}), \nabla(\nabla n(\mathbf{r})), \ldots]. \tag{5.22}$$

Apart from the fact that the exact form of the energy-functional is unknown, inclusion of density gradients makes the solution of the DFT equations rather difficult, and usually the *Ansatz* is made that the exchange correlation energy leads to an exchange correlation potential depending on the value of the density in $\mathbf{r}$ only and not on its gradients – this is the *local density approximation* (LDA):

$$E_{xc} = \int d^3r \, \varepsilon_{xc}[n(\mathbf{r})]n(\mathbf{r}) \tag{5.23}$$

where $\varepsilon_{xc}[n]$ is the exchange correlation energy per particle of an homogeneous electron gas at density n. The local density approximation is exact for an homogeneous electron gas, so it works well for systems in which the electron density does not vary too rapidly. We shall briefly discuss the various forms used for the exchange correlation energy density in the local density approximation, $\varepsilon_{xc}[n(\mathbf{r})]$, and refer to the literature for more details [4, 8, 9].

The exchange effects (denoted by the subscript 'x') are usually included in a term based on calculations for the homogeneous electron gas [10] giving the following form for the exchange energy in density functional theory:

$$\varepsilon_x[n(\mathbf{r})] = \text{Const.} \times n^{1/3}(\mathbf{r}) \tag{5.24}$$

which we have already encountered at the end of Section 4.5.1. The value for the constant is found as $-(3/4)(3/\pi)^{1/3}$.

For open-shell systems the spin-up and -down densities n_+ and n_- are usually taken into account as two independent densities in the exchange correlation energy according to a natural extension of the DFT formalism [4]. In local density approximation (now called local spin density approximation), the exchange is given as

$$E_{\mathrm{x}}[n_+, n_-] = -\mathrm{Const.} \int \mathrm{d}^3 r \; [n_+^{4/3}(\mathbf{r}) + n_-^{4/3}(\mathbf{r})], \tag{5.25}$$

with Const. $= (3/2)(3/4\pi)^{1/3}$ in accordance with the closed-shell prefactor in (5.24), as can be checked by putting $n_+ = n_- = n/2$. As is to be expected for an exchange coupling, this expression contains interactions between parallel spin pairs only.

In addition to exchange, there is a contribution from the dynamic correlation effects (due to the Coulomb interaction between the electrons) present in the exchange correlation potential, and several local density parametrisations of this interaction have been proposed. A successful one is a parametrised version of the correlation energy obtained in quantum Monte Carlo simulations of the homogeneous electron gas at different densities [11, 9]. Other parametrisations have been presented by von Barth and Hedin [12], and Gunnarson and Lundqvist [13]. These dynamic correlations represent couplings between both parallel and opposite spin pairs.

In calculations of the electronic structure, the DFT–LDA approach has turned out very successfully. In some systems, however, it leads to noticeable deviations or even failures – for examples some stable negative ions such as H^-, O^- and F^- are predicted to be unstable. Many improvements on LDA have therefore been proposed. One of these consists of including gradients of the density in the exchange correlation functional (we will come back to this in the second part of the next section), whose form is motivated by calculations taking many-electron effects into account [8].

Another approach focuses on the self-interaction present in the Hartree energy which contains Coulomb couplings between an electron and its own charge distribution. This overestimation of the electron–electron interaction should be cancelled by the exchange correlation term, which – in LDA – succeeds only partially (although in the hydrogen atom for example, 95% of the self-interaction is cancelled by the exchange correlation). It is possible to add these corrections afterward to the exchange correlation potential [9], but this introduces a dependence of the exchange correlation on the individual orbitals, ψ_k, instead of a dependence on the density only. Both the gradient-correction and self-interaction methods lead to important improvements in calculations of physical properties [4].

5.3 Exchange and correlation: a closer look

5.3.1 The adiabatic theorem and the normalisation conditions

In this section we consider exchange and correlation in more detail. We shall take into account the spin as well as the spatial coordinates. All spin-space coordinates $(\mathbf{r}_1, s_1; \ldots \mathbf{r}_N, s_N)$ are denoted by X. Let us first consider the exact energy-functional (of the spin-orbitals):

$$E_{\text{exact}} = \int \Psi^*_{\text{AS}}(X) \left(-\frac{1}{2} \sum_i \nabla_i^2 + V_{\text{ext}} + V_{ee} \right) \Psi_{\text{AS}}(X) \mathrm{d}X. \tag{5.26}$$

Here, $\Psi_{\text{AS}}(X)$ is a wave function which is antisymmetric in the $x_i = (\mathbf{r}_i, s_i)$, but not necessarily a Slater determinant. We compare the exact energy with the Kohn–Sham functional (which should also be exact for the correct exchange-correlation functional):

$$E_{\text{KS}} = -\sum_k \int \psi_k^*(x) \frac{1}{2} \nabla_k^2 \psi_k(x) \mathrm{d}x + \sum \int n(\mathbf{r}) V_{\text{ext}}(\mathbf{r}) \mathrm{d}x$$
$$+ \frac{1}{2} \int n(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} n(\mathbf{r}') \, \mathrm{d}^3 r \, \mathrm{d}^3 r' + E_{\text{xc}}[n]. \tag{5.27}$$

The terms related to $V_{\text{ext}}(\mathbf{r})$ are the same in both cases: the exchange and correlation term E_{xc} makes up for the difference in the kinetic energies and the difference between the exact Coulomb interaction and the Hartree approximation in the Kohn–Sham scheme.

We now try to connect the exact form to the Kohn–Sham picture in order to pinpoint this difference better. This is done in the *adiabatic connection* procedure, which works as follows. We first introduce a tunable electron–electron interaction

$$V_{\text{c},\lambda} = \sum_{ij} \frac{\lambda}{|\mathbf{r}_i - \mathbf{r}'_j|} = \lambda V_{\text{c}}, \tag{5.28}$$

where the subscript C stands for Coulomb and where V_{c} is identified with $V_{\text{c},\lambda=1}$.

Just as we did in Section 5.1.2, we split the many-body Hamiltonian into that of an homogeneous electron gas with interaction V_λ and the external potential:

$$H_\lambda = H_{0,\lambda} + \sum_i V_{\text{ext}}(\mathbf{r}_i) = (T + V_{\text{c},\lambda}) + \sum_i V_{\text{ext}}(\mathbf{r}_i), \tag{5.29}$$

and note that for *fixed* densities $n(\mathbf{r})$, the last term will always give the same contribution to the energy. Indeed, we minimise this Hamiltonian for such a fixed density:

$$E_\lambda[n] = \min_{\psi|n} \langle \Psi | H_{0,\lambda} | \Psi \rangle + \int V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \mathrm{d}^3 r = F_\lambda[n] + \int V_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \mathrm{d}^3 r, \tag{5.30}$$

where we have used the definition

$$F_\lambda[n] = \min_{\psi|n} \langle \Psi | H_{0,\lambda} | \Psi \rangle \tag{5.31}$$

The minimisation is carried out on the set of wave functions compatible with the given densities $n(\mathbf{r})$.

We now need a theorem that plays an important role in the quantum molecular dynamics method (see Chapter 9): the *Hellmann–Feynman theorem*. Here we shall prove this theorem for the simple case in which we have a Hamiltonian depending on a single parameter λ. The theorem tells us how the energy eigenvalues of a Hamiltonian H_λ, depending on a parameter λ, vary with λ. Differentiating the Schrödinger equation

$$H_\lambda |\psi_\lambda\rangle = E_\lambda |\psi_\lambda\rangle \tag{5.32}$$

with respect to λ we obtain (the prime indicates derivative with respect to λ):

$$H'_\lambda \| \psi_\lambda\rangle + H_\lambda \| \psi'_\lambda\rangle = E'_\lambda |\psi_\lambda\rangle + E_\lambda |\psi'_\lambda\rangle. \tag{5.33}$$

Taking the inner product from the left with $\langle \psi_\lambda |$ and using the Hermitian conjugate of (5.32), we see that

$$\frac{\mathrm{d}E_\lambda}{\mathrm{d}\lambda} = \frac{\langle \psi_\lambda | \mathrm{d}H_\lambda / \mathrm{d}\lambda | \psi_\lambda \rangle}{\langle \psi_\lambda | \psi_\lambda \rangle}. \tag{5.34}$$

We can write the derivative of F_λ from the Hellmann–Feynman theorem, by realising that, since $|\psi_\lambda\rangle$ is the variational ground state of F_λ, it must be the lowest eigenstate of $H_{0,\lambda}$. We then obtain

$$\frac{\mathrm{d}F_\lambda}{\mathrm{d}\lambda} = \langle \Psi_\lambda | V_\mathrm{c} | \Psi_\lambda \rangle. \tag{5.35}$$

From this, and from the fact that for $\lambda = 0$ we have trivial, noninteracting electron gas, we have

$$F_{\lambda=1}[n] = T_\mathrm{KS}[n] + \int_0^1 \langle \psi_\lambda | V_\mathrm{c} | \Psi_\lambda \rangle \mathrm{d}\lambda. \tag{5.36}$$

We now find the exchange correlation potential as the difference between the interacting and noninteracting electron gas including the Hartree energy E_H:

$$\begin{aligned} E_\mathrm{xc} &= F_{\lambda=1}[n] - T_{KS}[n] - \sum \frac{1}{2} \int n(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{r}'|} n(\mathbf{r}')\, \mathrm{d}^3 r\, \mathrm{d}^3 r' \\ &= \int_0^1 \langle \psi_\lambda | V_\mathrm{c} | \Psi_\lambda \rangle \mathrm{d}\lambda - E_\mathrm{H}. \end{aligned} \tag{5.37}$$

The main point of the derivation is that in (5.36), which holds for the interacting gas, the kinetic energy is that of the noninteracting gas; therefore, we find the exchange correlation correction only in terms of the Coulomb interaction. For $\lambda = 0$, the XC correction term is nonzero as the Hartree energy does not take the antisymmetry of the full many-body wave function into account: it is the exchange-only part of the correction.

There is another fruitful way of looking at the exchange correlation term, which is related to the discussion in Section 5.1.2. There we considered the probability density for finding the particles 1 and 2 with coordinates x and x' respectively:

$$P(x,x') = \int |\Psi(x,x',x_3,\ldots,x_N)|^2 \mathrm{d}x_3 \ldots \mathrm{d}x_N. \tag{5.38}$$

We now use this definition for a general wave function (not necessarily a Slater determinant).

The single-particle density is given as

$$n(x) = N \int |\Psi(x,x_2,\ldots,x_N)|^2 \mathrm{d}x_2 \ldots \mathrm{d}x_N. \tag{5.39}$$

Integrating the single-particle density gives the number of particles:

$$\int n(x)\mathrm{d}x = N. \tag{5.40}$$

From the definition of $n(x)$ we immediately see that

$$N \int P(x,x')\mathrm{d}x' = n(x). \tag{5.41}$$

The reason for introducing these quantities is that they give insight in the exchange and correlation energies. To see this, consider the Coulomb energy:

$$E_\mathrm{c} = \frac{N(N-1)}{2} \int P(x,x') \frac{1}{|\mathbf{r}-\mathbf{r}'|} \,\mathrm{d}x\,\mathrm{d}x' \tag{5.42}$$

(the prefactor counts the number of particle pairs). We now define the *exchange correlation hole*, $n_\mathrm{xc}(x,x')$, through

$$N(N-1)P(x,x') = n(x)n(x') + n(x)n_{xc}(x,x'). \tag{5.43}$$

The exchange correlation hole indicates how the actual distribution of a second electron, given a first electron at x, deviates from the average density. Then we can write

$$E_\mathrm{c} = E_\mathrm{H} + \frac{1}{2} \int n(x)n_\mathrm{xc}(x,x') \frac{1}{|\mathbf{r}-\mathbf{r}'|} \,\mathrm{d}x\,\mathrm{d}x'. \tag{5.44}$$

Note that the second term can be identified with the exchange correlation energy.

The most important consequence of this is that we can derive some properties of the exchange correlation hole, which any exchange correlation energy should satisfy. The first of these properties follows from the normalisation of P:

$$\int P(x,x') \,\mathrm{d}x\,\mathrm{d}x' = 1 \tag{5.45}$$

which follows directly from the normalisation of the wave function. Furthermore,

$$\int n(x)\mathrm{d}x = N \tag{5.46}$$

for the same reason. Integrating Eq. (5.43) now over $\mathrm{d}x'$ (this actually denotes an integration over $\mathbf{r}'$ and a sum over s'), we obtain the result

$$(N-1)n(x) = Nn(x) + n(x)\int n_{\mathrm{xc}}(x,x')\,\mathrm{d}x'; \tag{5.47}$$

in other words,

$$\int n_{\mathrm{xc}}(x,x')\mathrm{d}x' = -1. \tag{5.48}$$

Realising that the second term in Eq. (5.44) is the exchange correlation correction to the Coulomb energy, we see that this correction can be described in terms of a charge distribution which carries a positive unit charge: this is the exchange correlation hole n_{xc}.

Now let us return to the Hartree–Fock approximation. There we considered a Slater determinant containing all exchange effects. If we apply the above analysis to a Slater determinant we obtain an exchange hole (Coulomb correlations are absent in this case) which integrates up to a charge -1 (that is, a positive hole), *irrespective of the strength* λ *of the Coulomb interaction*. Therefore we conclude that the exchange hole adds up to -1 and, supposing that the exchange correlation hole is the sum of an exchange and a correlation contribution, the correlation hole must add up to 0.

Let us summarise the results obtained so far. The first is that we can remove the exchange and correlation contribution to the kinetic energy from the description by applying the adiabatic connection formula. The price we have to pay is that we have to integrate the Coulombic term due to exchange and correlation over the interaction strength λ. The second result is that this contribution can be described in terms of an exchange and a correlation hole, the first of which integrates up to -1 and the second integrates to 0.

5.3.2 The generalised gradient approximation

We can now understand the success of LDA: the exchange and correlation holes are taken from very accurate quantum Monte Carlo results for the homogeneous electron gas and therefore they satisfy the two normalisation conditions for exchange and correlation just described.

We can now also describe how a gradient expansion can be constructed: we must take into account isotropy conditions and then make sure that the exchange and the correlation hole satisfy their respective normalisation conditions. This scheme has been carried out by several groups, and some well known functionals are those of Perdew and Wang of 1986 [14, 15] and 1991 [16] (respectively PW86 and PW91), and of Becke [17], Lee, Yang and Parr [18] (LYP) and Perdew, Burke and Enzerhof [19, 20]. These exchange correlation functionals go by the name of generalised gradient approximations (GGAs).

In general, GGA improves on LDA for the quantities which are already successfully treated in LDA: total energies and hence binding energies, bond lengths and angles. Ionisation energies based on Kohn–Sham energy eigenvalues are approximately the same as for LDA. In general, LDA tends to over-estimate the correlation energy and underestimates the exchange energy; these are remedied to some extent in GGA, but as the two corrections are opposite, the net effect is not too spectacular. That does not mean that the improvement is not important: the GGA gives a more accurate description of the many-body electron system than LDA.

One major deficiency which is shared by GGA and LDA is the fact that the exchange correlation correction does not cancel the self-interaction present in the Hartree energy. This in particular affects the interpretation of the highest Kohn Sham energy as the ionisation energy of the system (see also Section 5.4.1).

5.3.3 Exact exchange

The problem with the known exchange functionals which are given as explicit functions of the density is the presence of self-interaction terms, a feature that was absent in the Hartree–Fock theory. It is possible to include the HF exchange term in the exchange correlation functional. This is justified in the so-called *optimized potential method* [21, 22] which leads to a Kohn–Sham picture where the exchange correlation functional is allowed to depend explicitly on the orbitals rather than on the density. The advantage of having no self-interaction left is counteracted by a less favourable scaling behaviour: just as in the HF theory, we must calculate and sum over two-electron integrals which makes this method rather time-consuming.

Finally, hybrid functionals combine exact exchange with traditional functionals.

5.4 Beyond DFT: one- and two-particle excitations

5.4.1 One-particle theories: ionisation and electron addition energies

The DFT is designed to yield correct ground state energies for a many-body system. It is, however, not justifiable to interpret Kohn–Sham energies as energy levels which can be detected in a spectroscopy experiment. An exception must be made for the highest occupied level, which gives the correct ionisation potential in exact DFT. To see that this is indeed the case, note that if one of the electrons (we take this to be electron number N) of a neutral system is moved very far away from all the nuclei in the system, the *exact* ground state wave function for the N electrons can be written as

$$\psi_N(\mathbf{r}_1, \ldots, \mathbf{r}_N) = \psi_{N-1}(\mathbf{r}_1, \ldots, \mathbf{r}_{N-1})\varphi(\mathbf{r}_N), \tag{5.49}$$

where this form is justified by the notion that at the large distance between particle N and its partners, any correlation between them has disappeared. Note that $\psi_{N-1}(\mathbf{r}_1, \ldots, \mathbf{r}_{N-1})$ is the normalised ground state wave function of the $N-1$ particles close to the nuclei, as the perturbation due to particle N can be neglected.

The Hamiltonian for the N-particle system can be written as [23]

$$H(N) = H(N-1) + \frac{p_N^2}{2m} + V_{\text{ext}}(\mathbf{r}_N) + \sum_{j=1}^{N-1} \frac{1}{|\mathbf{r}_j - \mathbf{r}_N|}, \tag{5.50}$$

where $H(N-1)$ is the $(N-1)$-particle Hamiltonian. Writing up the Schrödinger equation for the N electrons, using the wave function (5.49) and using the fact that the first term on the right hand side of that equation represents the $(N-1)$-particle ground state, we obtain an equation for φ:

$$\left[E_{N-1}^{\text{GS}} + \frac{p_N^2}{2m} + V_{\text{ext}}(\mathbf{r}_N) + \left\langle \psi_{N-1} \left| \sum_{j=1}^{N-1} \frac{1}{|\mathbf{r}_j - \mathbf{r}_N|} \right| \psi_{N-1} \right\rangle \right] \varphi(\mathbf{r}_N) = E_N^{\text{GS}} \varphi(\mathbf{r}_N). \tag{5.51}$$

The asymptotic (large r) behaviour of this equation is exactly the same as that of the Kohn–Sham equation (which also describes an electron far away from a localised charge distribution with net charge $+1$), and this can only be the case when the 'energy' eigenvalue of the Kohn–Sham equation is the same as $E_N^{\text{GS}} - E_{N-1}^{\text{GS}}$ [24].

This is a very interesting result when it is combined with Janak's theorem [25] which says that the highest occupied orbital energy gives the chemical potential (see Problem 5.4). In DFT, we can fill the orbitals partially by calculating the density with a fractional filling factor f_j:

$$n(\mathbf{r}) = \sum_j f_j |\psi(\mathbf{r})|^2. \tag{5.52}$$

This shows that we can really take an infinitesimal differential of the total energy (by varying f_N) with respect to the charge in the highest occupied orbital, which is the proper definition of the chemical potential. From the fact that the chemical potential and the ionisation energy are both given as the highest occupied Kohn–Sham eigenvalue, we see that the discrete derivative of the total energy with respect to the charge in the highest occupied level must be equal to the continuous derivative.

Perdew *et al.* [26] have argued that the derivative is constant for any fractional occupation of the highest occupied level, but their reasoning can be criticised because they impose this property in their form of the energy-functional (based on a density operator form), which need not describe the pure-state functional of DFT [27].

The property we have just derived – the Kohn–Sham energy of the highest occupied level gives us the ionisation energy – is satisfied very well for extended systems, but poorly for molecules, where the highest occupied Kohn–Sham energy (called the highest occupied molecular orbital, or HOMO) is generally found a few eV *above* the experimental value. Hartree–Fock usually gives a much better value. The reason why DFT fails so badly in practice lies in the poor asymptotic behaviour of the available exchange-correlation potentials, which, among other imperfections, do not cancel the self-interaction and hence give an incorrect asymptotic form of the Kohn–Sham potential. In our derivation, this asymptotic behaviour played a crucial role. The fact that we do not have the exact exchange correlation functional at our disposal therefore is a serious handicap in describing the spectra of atoms and molecules.

There is a way around this: given the fact that DFT is very good at calculating ground state energies, we can perform two calculations: one for N electrons, and one for $N-1$ (for the electron addition energies, the second calculation would be performed for $N+1$ electrons). The difference in the total energies then gives the ionisation (or electron addition) energy. Instead of these two energies, it is also possible to do one calculation at half filling of the highest occupied (or lowest unoccupied) level. The Kohn–Sham energy of that level is the derivative of the total energy with respect to the charge, so that we can predict the ionisation energy from

$$E^{\mathrm{GS}}(N) - E^{\mathrm{GS}}(N-1) = \left[\frac{\partial E_{\mathrm{tot}}}{\partial N}\right]_{N+1/2} = \varepsilon_N^{\mathrm{KS}}(N+1/2). \tag{5.53}$$

A similar procedure gives the electron addition energy. This method is known as *delta-SCF*.

5.4.2 General theories for excitation energies

Looking at what causes a system which is in the ground state to go to an excited state, we conclude that there should always be some time-dependent perturbation to the Hamiltonian which is responsible for this. Therefore, we should consider the response of the system to an external, time-dependent perturbation. The standard approach is to consider the response to a monochromatic perturbation with frequence ω. However, if the response of the system to a perturbation is faster than the typical period of the perturbation, we may consider a time-independent approach.

An electron which has been excited to a higher energy level will return to its ground state orbital after some time. This finite lifetime gives rise to a finite width of the energy spectrum, according to the time–energy uncertainty relation. Therefore, we can no longer speak of a discrete energy level, but we can still find a fingerprint of the spectrum in quantities such as the macroscopic dielectric function, which is the long-wavelength limit of the microscopic function $\varepsilon(\mathbf{r}, \mathbf{r}, \omega)$. This will exhibit peaks as a function of ω whose centres can be viewed as energy levels, and the widths as lifetimes.

Experimentally, excited states are studied by using spectroscopy techniques. In *direct* photo-emission, an incident photon excites an electron to sufficiently high energy that it can leave the system (ionisation). In *inverse* photo-emission we send an electron into the material to occupy an unoccupied state, causing emission of a photon whose energy is detected (electron addition). In absorption spectroscopy, the electron or photon that is sent into the system is also detected when it leaves the system. It may meanwhile have changed its energy by interaction with another electron in the material which is excited to a higher state.

The first two processes, ionisation and electron addition, are called *one-particle* processes; the third is a *two-particle* process. The two-particle character arises because, when an electron is excited in a system, it leaves a (positive) hole behind. If the electron remains in the system, it interacts with the hole, and in particular it may form an *exciton*: a bound state of the particle–hole system. We shall briefly describe the analysis for one-particle processes, and then review two-particle methods.

In the previous subsection, the problem of finding the ionisation energy was addressed. In general, when performing spectroscopy experiments, levels other than the highest one may be excited. Of course, one could try to use a generalised delta-SCF procedure for these, but this is difficult because for a band structure calculation, we would need many calculations as each $\mathbf{k}$-vector in the Brillouin zone has its own particular excitation. Another problem is that for a band state, DFT differs essentially from Hartree–Fock, which allows for calculating excited states: the HF orbital energies can be interpreted as excitation energies according to Koopman's theorem (which only holds for the highest band in DFT, see above). This theorem is based on the assumption that the orbitals do not relax when the configuration changes by emptying full, or filling empty levels. This approximation fails miserably in solids, for example in diamond, where the band gap is in HF predicted to be 15 eV, more than twice the experimental gap of about 7 eV.

What is missing from the description of a ground state system, is the fact that an electron added to the system does not feel the pure Coulomb interaction from the ground state charge distribution: the resident electrons will re-order in the presence of the visiting electron, and tend to *screen* the effect of the Coulomb interaction. A many-body theory for band structure takes these effects into account; HF and DFT do not.

Such a many-body theory was formulated by Hedin in 1965 [28], for reviews see Refs. [29, 30]. We shall not go into the details of the many-body theory behind this approach, but consider a particular, relatively simple form, the COHSEX approximation in some detail (we shall explain the name COHSEX below). This approximation can be understood quite well without going into the formal theory.

Suppose we put an extra unit charge into the system. This charge will occupy some state with orbital wave function $\psi(\mathbf{r})$. We could describe the interaction between this electron and the resident electrons by Hartree–Fock terms, i.e. a Coulomb interaction and an exchange interaction. However, although exchange is treated correctly (apart from neglecting screening effects, see below), the Coulomb interaction will push the resident electrons away from the visitor and thereby lower the interaction between visitor and residents.

Let us first neglect screening. The electrostatic energy is then given by

$$E_{\mathrm{ES}}(\mathbf{r}) = \int n(\mathbf{r}')v(\mathbf{r},\mathbf{r}')|\psi(\mathbf{r})|^2 \mathrm{d}^3r\, \mathrm{d}^3r', \tag{5.54}$$

where the interaction $v(\mathbf{r},\mathbf{r}')$ is the 'bare' Coulomb interaction potential $v(\mathbf{r},\mathbf{r}') = 1/|\mathbf{r}-\mathbf{r}'|$. Screening can be viewed from two different standpoints. The first is to consider the change Δn in charge distribution due to the presence of the new electron. The second view is to take for the potential felt by this electron a screened potential $w(r)$ which falls off more rapidly than the bare potential. Obviously, the two are connected.

For the correction to the energy due to the change $\Delta n(\mathbf{r})$ in the charge distribution we write:

$$\Delta E = \int \Delta n(\mathbf{r}')v(\mathbf{r},\mathbf{r}')|\psi(\mathbf{r})|^2 \mathrm{d}^3r\, \mathrm{d}^3r'. \tag{5.55}$$

However, this result is wrong, because the response Δn to the test charge is proportional to that charge! Therefore, if we integrate the energy up over the extra charge put into the system, we get a prefactor of $1/2$:

$$\Delta E = \frac{1}{2}\int \Delta n(\mathbf{r}')v(\mathbf{r},\mathbf{r}')|\psi(\mathbf{r})|^2\, \mathrm{d}^3r\, \mathrm{d}^3r'. \tag{5.56}$$

In order to get a handle on the screened potential w, we note that it is defined as the potential measured at $\mathbf{r}'$ given the fact that there is a test point charge at $\mathbf{r}$. We therefore may write:

$$\begin{aligned} w(\mathbf{r}',\mathbf{r}) &= \int \delta(\mathbf{r}-\mathbf{r}'')v(\mathbf{r}'',\mathbf{r}')\mathrm{d}^3r'' + \int \Delta n(\mathbf{r}''|\mathbf{r})v(\mathbf{r}'',\mathbf{r}')\mathrm{d}^3r'' \\ &= v(\mathbf{r},\mathbf{r}') + \int \Delta n(\mathbf{r}''|\mathbf{r})v(\mathbf{r}'',\mathbf{r}')\, \mathrm{d}^3r''. \end{aligned} \tag{5.57}$$

Here, $\Delta n(\mathbf{r}'|\mathbf{r})$ is the change in the charge density at $\mathbf{r}'$ due to a unit test charge placed at $\mathbf{r}$. The induced charge density $\Delta n(\mathbf{r})$ due to a charge distribution $|\psi(\mathbf{r})|^2$

is given as the integral of the induced charge $\Delta n(\mathbf{r}''|\mathbf{r})$ over $\mathbf{r}$, weighted by $|\psi(\mathbf{r})|^2$. Putting these results back into Eq. (5.56) leads to the formal expression

$$\Delta E = \frac{1}{2}\int \delta(\mathbf{r}-\mathbf{r}')[w(\mathbf{r},\mathbf{r}') - v(\mathbf{r},\mathbf{r}')]|\psi(\mathbf{r})|^2\, \mathrm{d}^3r\, \mathrm{d}^3r'. \tag{5.58}$$

If we take the functional derivative of this expression with respect to $\psi(\mathbf{r})$, we obtain a term

$$V\psi(\mathbf{r}) = \frac{1}{2}\int \delta(\mathbf{r}-\mathbf{r}')[w(\mathbf{r},\mathbf{r}') - v(\mathbf{r},\mathbf{r}')]\, \mathrm{d}^3r'\, \psi(\mathbf{r}) \tag{5.59}$$

in the one-particle Schrödinger equation. Exchange is already treated correctly, so in the exchange term, we can simply replace the bare Coulomb interaction by the screened interaction. We see that the correction boils down to taking into account the COulomb Hole and Screened EXchange – hence the name COHSEX.

Now there is still something missing: we do not know the screened interaction potential $w(\mathbf{r},\mathbf{r}')$. This can however be found in a so-called random phase approximation (RPA) scheme, which is based on perturbation theory. It works as follows. We place a test charge at position $\mathbf{r}$. As we have seen above, this test charge generates a change Δn of the resident charge distribution, and the bare potential $v(\mathbf{r},\mathbf{r}')$ is replaced by the screened potential $w(\mathbf{r},\mathbf{r}')$. The relation between the two is usually formulated in terms of the dielectric constant. This is defined as:

$$v(\mathbf{r},\mathbf{r}') = \int \varepsilon(\mathbf{r}',\mathbf{r}'')w(\mathbf{r},\mathbf{r}'')\, \mathrm{d}^3r''. \tag{5.60}$$

We can therefore write for the screening correction

$$\begin{aligned} w(\mathbf{r},\mathbf{r}') - v(\mathbf{r},\mathbf{r}') &= \int [\delta(\mathbf{r}',\mathbf{r}'') - \varepsilon(\mathbf{r}',\mathbf{r}'')]w(\mathbf{r},\mathbf{r}'')\, \mathrm{d}^3r'' \\ &= \int \Delta n(\mathbf{r}''|\mathbf{r})v(\mathbf{r}',\mathbf{r}'')\mathrm{d}^3r'', \end{aligned} \tag{5.61}$$

where $\Delta n(\mathbf{r}''|\mathbf{r})$ denotes a change of the density at $\mathbf{r}''$ due to a unit point charge being placed at $\mathbf{r}$.

Now we view the effect of this point charge at $\mathbf{r}$ as a perturbing potential w. The lowest order correction to the occupied orbital j in stationary perturbation theory is given by [31]

$$\Delta\psi_j(\mathbf{r}') = \sum_{k \text{ unocc.}} \frac{\langle\psi_k|w|\psi_j\rangle}{E_j - E_k}\psi_k(\mathbf{r}'). \tag{5.62}$$

The total change in the density is therefore given by

$$\Delta n(\mathbf{r}') = 2 \sum_{j \text{ occ.}} \psi_j^*(\mathbf{r}')\Delta\psi_j(\mathbf{r}')$$
$$= 2{\sum_{j,k}}' \int \frac{\psi_j^*(\mathbf{r}')\psi_k(\mathbf{r}')\psi_k^*(\mathbf{r}'')\psi_j(\mathbf{r}'')}{E_j - E_k} w(\mathbf{r}', \mathbf{r})\, \mathrm{d}^3 r'', \tag{5.63}$$

where the prime with the sum indicates that the index j runs over occupied, and k over unoccupied levels. Putting this back into the rightmost term of Eq. (5.61) and using the equality between the second and third expression in this equation yields

$$\varepsilon(\mathbf{r}', \mathbf{r}) - \delta(\mathbf{r}', \mathbf{r}) = 2{\sum_{j,k}}' \int\int \frac{\psi_j(\mathbf{r})\psi_k^*(\mathbf{r})\psi_k^*(\mathbf{r}'')\psi_j(\mathbf{r}'')}{(E_j - E_k)|\mathbf{r} - \mathbf{r}'|}\, \mathrm{d}^3 r''. \tag{5.64}$$

In this derivation we have assumed that the effects of the new electron on the resident one can be completely described in terms of the Hartree term in the Hamiltonian. This is known as the *random phase approximation* [32].

Hybertsen and Louie have implemented the full GW approximation into an LDA framework [33, 34], and obtained energy spectra with excellent agreement with experiment. The static COHSEX approximation is only a first step in this procedure. It is possible to replace the relation (5.60) by a local one:

$$v(\mathbf{r}, \mathbf{r}') = w(\mathbf{r}, \mathbf{r}')\varepsilon(\mathbf{r}, \mathbf{r}'), \tag{5.65}$$

which is sometimes done for convenience. The detailed structure overlooked in this approximation is denoted as *local field effects*. From the work of Hybertsen and Louie it is clear that local field effects have a major impact on the energy spectra.

We see that a particle which is added to the system will influence the behaviour of the other particles. If we could switch off the interaction between the particles, the newly added particles would occupy sharp energy levels, and the new particle on its own would completely determine the new level. Landau [35] analysed the many-body behaviour of liquid helium-3 and argued that if we had a knob with which we could tune the interactions, the spectrum would change in a *continuous* way. That is, for no interaction, the spectrum consists of a series of delta-functions, which start to broaden and shift when the interactions are switched on. The corresponding excitations involve, as we have seen, the presence of a new particle (or, in the case of two-particle problems, a particle occupying a new state), accompanied by a slight change of the orbitals of the other particles. This excitation is called a *quasi-particle*. Quasi-particle excitations can be analysed in terms of many-body Green's functions [36].

5.4.3 Two-particle effects

A two-particle description within the many-body theory of Green's functions can be formulated: it is known as the *Bethe–Salpeter* theory. Implementation of this is possible but generally demanding – for a review see Ref. [37]. Another approach which potentially describes any type of excitation of a many-body system in the presence of a time-dependent field is *time-dependent density functional theory* (TDDFT) [38–40]. The formalism of this theory is analogous to that of plain DFT, and the analogue of the DFT Hohenberg–Kohn theorem in TDDFT is the *Runge–Gross* theorem. This reads:

> Two densities $\rho(\mathbf{r},t)$ and $\rho'(\mathbf{r},t)$ evolving from a common initial state $\psi(\mathbf{R},t=0)$ $[\mathbf{R}=(\mathbf{r}_1,\mathbf{r}_2,\ldots,\mathbf{r}_N)]$ under the influence of two external potentials $v(\mathbf{r},t)$ and $v'(\mathbf{r},t)$ are always different provided these potentials differ by more than a purely time-dependent function
> $$v'(\mathbf{r},t) \neq v(\mathbf{r},t) + c(t). \tag{5.66}$$

The presence of the uniform function $c(t)$ in this last condition is related to the 'gauge invariance': multiplying $\psi(\mathbf{r},t)$ by a factor $\exp[-\mathrm{i}C(t)/\hbar]$ solves the time-dependent Schrödinger equation with a potential shifted by $c(t)=\dot{C}(t)$. This is easily verified.

A time-dependent Kohn–Sham formulation can be derived from this theorem. This formulation gives the time-evolution of single-particle orbitals which generate the same density as the full many-body problem. These orbitals evolve according to a time-dependent Schrödinger equation:

$$\mathrm{i}\frac{\partial\psi_k(\mathbf{r},t)}{\partial t} = \left[-\frac{1}{2}\nabla^2 + V_{\mathrm{ext}}(\mathbf{r},t) + V_{\mathrm{H}}(\mathbf{r},t) + V_{\mathrm{xc}}(\mathbf{r},t)\right]\psi_k(\mathbf{r},t)\mathrm{for}k=1,\ldots,N. \tag{5.67}$$

The density is now time-dependent: it is as usual given by

$$n(\mathbf{r},t) = \sum_{k=1}^{N} |\psi_k(\mathbf{r},t)|^2. \tag{5.68}$$

The Hartree and exchange-correlation potentials V_{H} and V_{xc} are defined in terms of the time-dependent density using the same expressions as in static DFT. Note, however, that an exchange-correlation potential that works in static DFT is not guaranteed to work in TDDFT. In fact this is the greatest weakness of TDDFT at this moment: it is as yet unclear which are the reliable approximations to this potential.

Technically, the solution of the time-dependent Kohn–Sham equations can be carried out in a Crank–Nicholson or in a split-operator scheme (see Appendix 7.2). The application of these schemes is however slightly tricky [41]. The reason is that in a proper Crank–Nicholson scheme, we use the Hamiltonian operator evaluated at $t + h/2$, where h is the time step in going from t to $t + h$. As the Hamiltonian depends on the solutions $\psi_k(t+h/2)$ which are not yet known, we must first estimate $\psi_k(t + h)$ using H evaluated at t ($\hbar = 1$; do not confuse with the time step h):

$$\tilde{\psi}_k(t + h) = \frac{1 + \mathrm{i}hH(t)/2}{1 - \mathrm{i}hH(t)/2}\psi_k(t). \tag{5.69}$$

Using the $\tilde{\psi}_k(t + h)$, we evaluate $\tilde{H}(t + h)$. Then we again perform a Crank–Nicholson step where we use the mean of $H(t)$ and $\tilde{H}(t + h)$.

In the split-operator scheme, the solution to the fact that the orbitals are unknown at $t + h/2$ can be solved in an elegant way [41]. The scheme brings us from t to $t + h$ by applying the following operation:

$$\psi_k(t + h) = \exp(-\mathrm{i}T/2)\exp[-\mathrm{i}V(t + h/2)]\exp(-\mathrm{i}T/2)\psi_k(t), \tag{5.70}$$

where T is the kinetic, and V the potential energy. In order to perform this step, we need to Fourier-transform back and forth between the momentum and direct-space representations where the operators occurring in the exponentials are diagonal:

$$\begin{gathered}\psi_k(\mathbf{r}, t) \xrightarrow{\text{FFT}} \psi_k(\mathbf{p}, t) \xrightarrow{\times\exp[-\mathrm{i}T/2)} \psi'_k(\mathbf{p}, t) \xrightarrow{\text{FFT}} \psi'_k(\mathbf{r}, t) \xrightarrow{\times\exp[-\mathrm{i}V(t+h/2)]} \\ \psi''_k(\mathbf{r}, t) \xrightarrow{\text{FFT}} \psi''(\mathbf{p}, t) \xrightarrow{\times\exp(-\mathrm{i}T/2)} \psi'''_k(\mathbf{p}, t) \xrightarrow{\text{FFT}} \psi'''_k(\mathbf{r}, t) = \psi_k(t + h).\end{gathered} \tag{5.71}$$

The nice property of applying the second operator ($\times\exp[-\mathrm{i}V(t + h/2)]$) is that it does *not* change the density, as it represents just a phase factor in real space. Therefore we can just take the orbitals ψ'_k to evaluate the potential in this procedure. This implies that we already have ψ_k at the half-integer steps at our disposal. Furthermore, we can glue the last stage of this procedure onto the first stage of the next step, at the expense of not having the ψ_k at our disposal at the integer time steps.

A particularly nice sample application of TDDFT is the description of *higher harmonic generation* in helium [42], which describes the generation of higher harmonics in the response to monochromatic light of high intensity [43]. Generally, TDDFT is a very useful tool for calculating dynamic response functions (frequency-dependent polarisabilities) [44].

5.5 A density functional program for the helium atom

In this section we describe the construction of a program for the calculation of the ground state of the helium atom within the local density approximation. As the two electrons occupy the 1s-orbital, the density and hence the Hartree potential are radially symmetric and we exploit this symmetry in spatial integrations. Instead of using basis functions, we solve the radial Schrödinger equation directly, just as we have done in the first chapter for calculating scattering cross sections. The program is set up in three steps. First, we use a simple integration algorithm and combine this with an interpolation routine in order to find the stationary states of hydrogen-like atoms. Second, a routine for obtaining the Hartree potential from the (radial) electronic density is added. At this point the results should compare with those obtained in the previous chapter using Gaussian basis functions. Finally, the exchange correlation potential is added and we have a fully self-consistent local density program.

5.5.1 Solving the radial equation

To solve the radial Schrödinger equation you can use the Numerov algorithm which is discussed in Appendix 7.1 and which has been used for the scattering program in Chapter 2. However, we also have to solve other differential equations and integrals, and in order to avoid complications we shall not require the $\mathcal{O}(h^6)$ accuracy of Numerov's algorithm – hence you can also use the simpler Verlet–Stoermer algorithm of Appendix A7.1. It is of course possible and recommended to use library routines throughout the program. For integrating the radial Schrödinger equation, a nonuniform grid is often used which is dense near the nucleus where the Coulomb potential diverges (see Problem 5.1). For the hydrogen atom, the radial equation for $l = 0$ reads (in atomic units)

$$\left[-\frac{1}{2}\nabla^2 - \frac{1}{r}\right]u(r) = Eu(r) \tag{5.72}$$

where $u(r)$ is given as $rR(r)$, $R(r)$ being the radial wave function. For the hydrogen atom we know that the solution for the ground state is given by $E = -0.5$ a.u. and $u(r) \propto r\mathrm{e}^{-r}$, and this enables us to test our programs. The energy eigenvalues can be found by integrating the radial Schrödinger equation from some large radius $r_{\max}$ inward to $r = 0$ and checking whether the solution vanishes there. The procedure is analogous to that described in Problem A4. You should first check whether for $E = -0.5$ a.u. the radial solution does indeed vanish at $r = 0$. Note that for the regular solutions $[u(0) = 0]$ we are looking for, the divergence of the potential near the origin causes no problems in the integration routine as long as it is not evaluated at $r = 0$.

For the starting values at $r_{\max}$ you can substitute the exact values $u(r_{\max}) = r_{\max}\exp(-r_{\max})$ and similarly for $u(r_{\max} - h)$, but it is also possible to take $u(r_{\max})$ equal to 0 and $u(r_{\max} - h)$ equal to a very small value. It is interesting to play around varying the starting conditions and the value of $r_{\max}$ in order to get a feeling for how the accuracy is affected by these.

To arrive at a program which determines the spectrum for you, you must couple the integration routine to a root-finding scheme and apply it to the value of u at the origin. Although it is in principle possible to solve for the energy derivative of u alongside the determination of u itself, we assume here that the integration routine does not provide energy derivatives. Therefore a library root-finding routine must not use the derivative and the same holds for one you write yourself. In the latter case, the secant method is appropriate; see Appendix A3. You will have to supply the boundaries between which the root must lie when using the program.

PROGRAMMING EXERCISE

Combine the integration routine and the root-finding routine into a method for finding the $l = 0$ states of a radial potential.

Check Test your program for the hydrogen atom.

5.5.2 Including the Hartree potential

We now describe an extension of the hydrogen program to the helium case, which implies having a nuclear potential $-2/r$ in the Hamiltonian and requires some treatment of the electron–electron interaction. In this section we take the latter into account in the same way as in Section 4.3.2, that is by a so-called Hartree potential which is the electrostatic potential generated by the charge distribution following from the wave function. This potential is given by

$$V_{\mathrm{H}}(\mathbf{r}) = \int \mathrm{d}^3 r' \; n_{\mathrm{s}}(\mathbf{r}') \frac{1}{|\mathbf{r} - \mathbf{r}'|}. \tag{5.73}$$

Here, n_{s} stands for the density of a *single* orbital – the total charge density is twice as large as a result of summation over the spin. The proper Hartree potential is therefore twice as large, but half of it consists of the self-interaction which we have subtracted off because this can easily be done for the helium case (see also the end of Section 4.3.1). Rather than solving for this potential by calculating the integral (5.73) directly, we shall find it by solving Poisson's equation:

$$\nabla^2 V_{\mathrm{H}}(\mathbf{r}) = -4\pi n_{\mathrm{s}}(\mathbf{r}). \tag{5.74}$$

Using the radial symmetry of the density and defining $U(r) = rV_{\mathrm{H}}(r)$, this equation reduces to the form

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2} U(r) = -4\pi r n_{\mathrm{s}}(r). \tag{5.75}$$

This is an ordinary second order differential equation which can be solved again using Verlet's algorithm (or a library routine). Note that it is necessary to normalise the radial wave function before integrating Poisson's equation! If you take for the normalisation

$$\int \mathrm{d}r\, u^2(r) = \int \mathrm{d}r\, r^2 R^2(r) = 1, \tag{5.76}$$

you have already included a factor 4π into the density (arising from the angular integrations) and the factor 4π in Poisson's equation drops out:

$$U''(r) = -\frac{u^2(r)}{r}. \tag{5.77}$$

We shall use the normalisation (5.76) throughout this section.

The solution of Eq. (5.77) contains two integration constants which have to be fixed by the boundary conditions. We take $U(0) = 0$ as the first boundary condition. Elementary electrostatics then leads to the second condition

$$V'_{\mathrm{H}}(r_{\mathrm{max}}) = \frac{q_{\mathrm{max}}}{r^2_{\mathrm{max}}}, \tag{5.78}$$

where q_{max} is the electron charge contained in a sphere of radius r_{max}:

$$q_{\mathrm{max}} = \int_0^{\mathrm{max}} \mathrm{d}r\, u^2(r). \tag{5.79}$$

For large r_{max}, q_{max} is the total electron charge. In that case we see from the asymptotic form of (5.78), using the fact that $U(r_{\mathrm{max}})$ is now constant as a function of r_{max}), that $U(r_{\mathrm{max}}) = q_{\mathrm{max}}$. When carrying out the numerical integration, we take for the first starting condition $U(0) = 0$. The second starting condition, for $U(h)$, is not known at the beginning – we take $U(h) = h$. As the solution $U(r) = \alpha r$ solves the homogeneous differential equation, $U''(r) = 0$, we can add this solution to the numerical solution found, with α taken such as to satisfy the end condition $U(r_{\mathrm{max}}) = q_{\mathrm{max}}$, without violating the starting condition $U(0) = 0$.

PROGRAMMING EXERCISE

Add an extra integration to your program which solves Eq. (5.77).

It is useful to check for correctness by using the hydrogen atom as an example. The normalised ground state density (in the sense of (5.76)), found at $E = -0.5$ a.u., is $4\mathrm{e}^{-2r}$ and we must solve

$$U''(r) = -4r\mathrm{e}^{-2r}, \tag{5.80}$$

with the boundary conditions $U(0) = 0$, $U(\infty) = 1$, so

$$U(r) = -(r+1)\mathrm{e}^{-2r} + 1. \tag{5.81}$$

Check Check whether your program produces these results

The next step is to make the program self-consistent. This is done by adding the Hartree potential to the nuclear potential and solving for the eigenstate again. You repeat this process until the energy does not change appreciably between subsequent steps. The total energy is given by

$$E = 2\varepsilon - \int \mathrm{d}r\ V_{\mathrm{H}}(r)u^2(r). \tag{5.82}$$

The Hartree correction arises because the Hartree energy is quadratic in the density.

Check Try to reproduce the results for the helium Hartree–Fock calculation in Section 4.3.2. In fact, the present method is more accurate as the wave functions are not restricted to linear combinations of four Gaussians. For an integration step $h = 0.01$ (in the Verlet algorithm) you will find for the eigenvalue of the radial Schrödinger equation the value -0.923 a.u. and for the Hartree correction 1.0155 a.u., so that the total energy amounts to $E = -2.861$ a.u., in good agreement with the result obtained in the previous chapter. The experimental value is -2.903 a.u.

5.5.3 The local density exchange potential

The aim of the exercise has not yet been achieved: we must calculate the energy and eigenvalues in the density functional formalism within the local density approximation. Remember that in density functional theory, the density that gives rise to the Hartree potential is the *full* density $n(\mathbf{r})$, i.e. the density of the two electrons, and in the previous section we have subtracted off the self-interaction contribution, leading to a reduction by a factor of 2 of the Hartree potential. Multiplying the Hartree potential by a factor of 2 in the previous program yields very poor results and therefore we hope that the exchange potential will correct for the self-interaction. As we have noted above, a popular form of the local density exchange potential is the one based on a treatment of the exchange hole in a homogeneous electron gas and is given by

$$V_{\mathrm{x}}(\mathbf{r}) = \text{Const.} \times n^{1/3}(\mathbf{r}) \tag{5.83}$$

where the constant is given as

$$\text{Const.} = -\left(\frac{3}{\pi}\right)^{1/3}. \tag{5.84}$$

Here, again, the *full* density is to be taken in the right hand side of (5.83) and this is twice the single electron density arising from the radial Schrödinger equation, since we have two electrons. Therefore, in terms of the radial eigenfunctions u

normalised as in (5.76), our exchange potential reads

$$V_{\mathrm{x}}(\mathbf{r}) = -\left[\frac{3u^2(r)}{2\pi^2 r^2}\right]^{1/3} \tag{5.85}$$

which, for the s-states under consideration, depends only on the radial coordinate r. The total energy is given by

$$E = 2\varepsilon - \int \mathrm{d}r\, V_{\mathrm{H}}(r)u^2(r) + \frac{1}{2}\int \mathrm{d}r\, u^2(r)V_{\mathrm{x}}(r). \tag{5.86}$$

The extension of your program to a local density version is now straightforward: instead of adding only the Hartree potential to the nuclear attraction, you take twice this potential and add the exchange potential to it. The self-consistency loop remains unaltered.

PROGRAMMING EXERCISE

Extend your Hartree–Fock program to include the exchange potential.

Check If your program is correct, it should give the following values for the energies: $\varepsilon = -0.52$ and $E = -2.72$ a.u.

Obviously the result is inferior to Hartree–Fock as the exchange potential is included only in an approximate way. Improvement is possible by considering an exchange correlation potential based on an interpolation of quantum Monte Carlo results by Ceperley and Alder [45], and it yields a ground state energy of $E = -2.83$ a.u. [9] which is an important improvement with respect to -2.72, although it is still worse than the HF result of -2.86 a.u. Implementation of this is straightforward and will be done in Problem 5.6.

5.6 Applications and results

In numerous calculations for atoms, molecules and solids the DFT–LDA approach has been very successful. In this section we quote some results which have been taken from the review by Jones and Gunnarson [4].

The original applications were to the ground state properties of solids, and some typical results are shown in Table 5.1. Binding energies for atoms and molecules are often better than HF (Table 5.3); total energies are close to but a bit worse than HF (Table 5.2). Interpretation of the Kohn–Sham eigenvalues as excitation energies works surprisingly well in many solids, where the energy bands frequently agree with those measured in photo-emission for example (see Problem 5.4 and

Table 5.1. *Lattice constants and cohesive energies for diamond,* Si *and* Ge. *Atomic units are used.*

	Lattice constant		Cohesion energy	
	DFT	Expt	DFT	Expt
Diamond	6.807	6.740	7.58	7.37
Si	10.30	10.26	4.84	4.64
Ge	10.69	10.68	4.02	3.85

Data taken from Ref. [4].

Table 5.2. *Energies in a.u. for various atoms.*

Atom	HF	DFT	Expt.
Li	−7.433	−7.353	−7.479
C	−37.702	−37.479	−37.858
O	−74.858	−74.532	−75.113

Data taken from Ref. [4]

Table 5.3. *Binding energies in a.u. for diatomic molecules.*

Atom	HF	DFT	Expt.
H_2	3.64	4.91	4.75
C_2	0.79	7.19	6.32
O_2	1.28	7.54	5.22

Data taken from Ref. [4].

Section 5.3). But we should be cautious about interpreting ψ_k and ε_k as anything other than auxiliary quantities for constructing the ground state energy and density as explained extensively in that section. There are several examples where interpretation of ε_k as excitation energies goes drastically wrong: band gaps in semiconductors and insulators are almost invariably too small, and ionisation energies for atoms and molecules are usually way too small. The inclusion of self-interaction corrections, mentioned in the previous subsection, gives better results for these gaps, but remember that these corrections introduce dependence of the Hamiltonian on individual orbitals instead of the density only and are therefore incompatible with DFT. The best approach is to use many-body theories for calculating actual excitation energies.

Exercises

5.1 [C] Instead of the regular grid which was used in the helium program of Section 5.5, it is better to use a grid with a step size which grows from a very small value near the nucleus to larger values in the valence region, because the wave function will oscillate more rapidly near the nucleus as a result of the deep Coulomb potential. Consider a grid with grid points given by the following formula:

$$r_j = r_{\mathrm{p}}[\exp(j\delta) - 1], \quad j = 0, 1, \ldots, j_{\max}.$$

The grid point with $j = 0$ coincides with the nucleus and the grid runs up to a radius $r_{\max}$ which fixes the value of the prefactor r_{p} to

$$r_{\mathrm{p}} = r_{\max}/[\exp(j_{\max}\delta) - 1].$$

The grid is defined by the number of grid points $j_{\max}$, by the outermost point $r_{\max}$ and by the parameter δ which determines how much the grid constant near the nucleus differs from that near $r_{\max}$. All these three values must be specified and then the prefactor r_{p} can be determined.

(a) Show that, in terms of j, the radial Schrödinger equation

$$\frac{\mathrm{d}^2}{\mathrm{d}r^2}u(r) = [V(r) - E]u(r)$$

transforms into

$$\frac{\mathrm{d}^2}{\mathrm{d}j^2}u(j) - \delta\frac{\mathrm{d}}{\mathrm{d}j}u(j) = r_{\mathrm{p}}^2\delta^2\mathrm{e}^{2j\delta}[V(j) - E]u(j),$$

where $u(j) = u(r_j)$.

(b) Write a general integral $\int_0^{\max} f(r)\mathrm{d}r$ as an integral over j.

(c) [C] Transform all integrals and differential equation methods in the density functional program to the nonhomogeneous grid defined above. Compare the accuracies of the two versions.

(d) Show that the first derivative occurring in the radial Schrödinger equation in terms of j above can be transformed away by writing $u(j) = v(j)\exp(j\delta/2)$. Show that the resulting equation for v reads

$$\frac{\mathrm{d}^2}{\mathrm{d}j^2}v(j) - \frac{\delta^2}{4}v(j) = r_{\mathrm{p}}^2\delta^2\mathrm{e}^{2j\delta}[V(j) - E]v(j).$$

(e) [C] Numerov's algorithm (see Appendix A7.1) can be used for solving this differential equation. Try this out for the ground state of the hydrogen atom and show that the numerical error scales as $1/N^4$ as is expected (see Problem A3). Note that when the number of points is doubled, δ should be decreased by a factor of 2.

5.2 [C] The Hartree energy

$$E_{\mathrm{H}} = \frac{1}{2}\int \mathrm{d}^3r\,\mathrm{d}^3r'\,\frac{n(\mathbf{r}')n(\mathbf{r})}{|\mathbf{r}-\mathbf{r}'|}$$

overestimates the classical electrostatic energy of the electrons because it includes interactions of the electrons with themselves – these are the so-called self-interactions. In Hartree–Fock theory, this spurious effect is cancelled by the exchange energy. In density functional theory, the exchange correlation energy does not ensure this cancellation a priori and we can only hope that it cancels the self-interaction as much as possible. To see to what extent the exchange correlation potential succeeds in doing so, we consider the hydrogen atom in DFT (of course, DFT was designed for many-electron systems, but its use is not a priori restricted to systems containing more than one electron). In the hydrogen atom, we find a nonvanishing Hartree and exchange correlation energy, which can easily be evaluated with our DFT program for helium.

Change the nuclear charge back to $Z = 1$ and make sure that the density used in the Hartree and exchange correlation energies is evaluated for the single electron (i.e. not multiplied by 2 as in the helium case). Evaluate both energies for the exact solution of the hydrogen atom.

You should find that the exchange correlation energy compensates about 80% of the self-interaction. For better exchange correlation energies, a value of 96% can be found – see the following problem and Ref. [9]. See also Ref. [46] for more examples.

5.3 [C] The Slater exchange potential

$$V_{\mathrm{x}}(\mathbf{r}) = -\left(\frac{3}{\pi}\right)^{1/3} n^{1/3}(\mathbf{r})$$

is based on the exchange energy in a homogeneous electron gas [47]. It is quite a crude approximation, and a refinement can be made using quantum Monte Carlo results obtained by Ceperley and Alder [9, 11, 45]. This leads to a parametrised correlation energy which should be *added to* the Slater term given above. The parametrisation is given in terms of the parameter r_s which is related to the density n according to

$$n = \frac{3}{4\pi r_s^3}.$$

The parametrisation is split into two parts: $r_s \geq 1$ and $r_s < 1$. We need an expression for the correlation energy parameter ε_{c} defined by

$$E_{\mathrm{c}} = \int \mathrm{d}^3r\, n(\mathbf{r})\varepsilon_{\mathrm{c}}(n)n(\mathbf{r}).$$

(a) Show that from this an expression for the correlation potential V_{c} can be derived according to

$$V_{\mathrm{c}}(r_s) = \left(1 - \frac{r_s}{3}\frac{\mathrm{d}}{\mathrm{d}r_s}\right)\varepsilon_{\mathrm{c}}(r_s).$$

Table 5.4. *Parameters for correlation energy*

	Unpolarised	Polarised
A	0.0311	0.01555
B	−0.048	−0.0269
C	0.0020	0.0014
D	−0.0116	−0.0108
γ	−0.1423	−0.0843
β_1	1.0529	1.3981
β_2	0.3334	0.2611

(b) [C] A parametrised form of ε_c is given by the following expressions. For $r_s \geq 1$ we have

$$\varepsilon_c = \gamma/(1 + \beta_1\sqrt{r_s} + \beta_2 r_s)$$

and for $r_s > 1$

$$\varepsilon_c = A \ln r_s + B + C r_s \ln r_s + D r_s.$$

From this, we obtain the following expressions for the correlation potential:

$$V_c(r_s) = \varepsilon_c \frac{1 + 7/6\beta_1\sqrt{r_s} + \beta_2 r_s}{1 + \beta_1\sqrt{r_s} + \beta_2 r_s}$$

for $r_s \geq 1$ and

$$V_c(r_s) = A \ln r_s + B - A/3 + \frac{2}{3} C r_s \ln r_s + (2D - C) r_s/3.$$

The values of the parameters A, B etc. depend on whether we are dealing with the polarised (all spins same z component) or unpolarised case. For both cases, the values are given in Table 5.4.

Use this parametrisation in your helium density functional theory program (unpolarised). You should find an energy $E = -2.83$ atomic units,to be compared with -2.72 without this correction.

(c) [C] Use the polarised parametrisation for the hydrogen program of the previous problem. You should find an energy $E = -0.478$ a.u.

(d) [C] It is also possible to combine the self-energy correction with the correlation energy. You should consult the paper by Perdew and Zunger [9], if you intend to do this. This results in an energy $E = -2.918$ a.u., which is only 0.015 a.u. off the experimental value.

5.4 In this problem, we consider a generalisation of Koopman's theorem (see Section 4.5.3) to the density functional formalism. To this end, we consider the spectrum $\{\varepsilon_i\}$ and the corresponding eigenstates of the Kohn–Sham Hamiltonian. We consider the chemical potential, which is found by removing a small amount of charge from the system. In practice this means that the highest level (which is level N) is not fully occupied. We usually calculate the density according to

$$n(\mathbf{r}) = \sum_{i=1}^{N} f_i |\psi_N(\mathbf{r})|^2.$$

The change in the density is realised by reducing the value f_N slightly:

$$f_N \to f_N - \delta f_N.$$

This induces a change in the density

$$\delta n(\mathbf{r}) = \delta f_N |\psi_N(\mathbf{r})|^2.$$

The total energy is calculated according to:

$$E(N) = \sum_{i=1}^{N} f_i \varepsilon_i - \int \frac{1}{2} \mathrm{d}^3 r \, \mathrm{d}^3 r' \, \frac{n(\mathbf{r}) n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} - \int \mathrm{d}^3 r \, n(\mathbf{r}) V_{\mathrm{xc}}(\mathbf{r}) + E_{\mathrm{xc}}[n].$$

The levels ε_i arise from taking the matrix elements $\langle \psi_i | H(N) | \psi_i \rangle$. As a result of the change in density, both the Hamiltonian occurring in these matrix elements and the remaining terms in the energy expression change.

We therefore have three contributions to the change in the total energy. First, the factor f_N in the sum over the energy levels changes; second, the potential for which the levels are calculated changes slightly; and third, the correction terms in the expression for the energy change.

Show that, to linear order in $\delta n(\mathbf{r})$, the combined effect of the change in the Hamiltonian matrix elements is precisely compensated by the change in the remaining terms in the energy expression so that we obtain

$$E(N) - E(N - \delta f_N) = \varepsilon_N \delta f_N$$

Hint: the change in the exchange correlation energy $E_{\mathrm{xc}}[n]$ is given by the expression

$$\delta E_{\mathrm{xc}}[n] = \int \mathrm{d}^3 r \, \delta n(\mathbf{r}) \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\mathbf{r})} = \int \mathrm{d}^3 r \, V_{\mathrm{xc}}[n](\mathbf{r}) \delta n(\mathbf{r}).$$

This proves Janak's theorem [25].

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